

SURF Guide

Free-Boundary-Aware Risk Surfaces for American Options

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Abstract

This document gives a step-by-step research plan for students working on American option pricing as a free-boundary problem. The project is deliberately focused on an application question: how to construct stable, financially consistent price, Greek, and early-exercise-boundary surfaces for American options under changing market regimes. Classical Crank–Nicolson finite differences with projected successive over-relaxation (CN/PSOR) will be used as the trusted numerical benchmark. Neural networks will be used only after the benchmark is stable, and primarily as constrained surrogates for fast risk surface evaluation.

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1 How to Use This Document

Students should treat this document as an execution guide, not only as reading material. Work through the stages in order:

1. reproduce the mathematical formulation;
2. implement and validate the classical solver;
3. generate application figures and diagnostics;
4. only then build neural surrogates.

Do not skip validation in order to get to neural networks more quickly. In this project, the scientific value comes from respecting the free-boundary structure.

2 Purpose of the Project

The project should not be framed as a generic comparison between finite difference methods and neural networks. That framing is too broad and too easy to make incremental. The intended paper is about a specific computational finance problem:

American option prices are not difficult to compute one by one in low dimension, but stable risk surfaces are harder: the early-exercise boundary creates non-smooth behaviour in Delta and Gamma, especially near the exercise frontier and near maturity.

The working title is:

**Free-Boundary-Aware Risk Surfaces for American Options:
Stable Greeks, Exercise Boundaries, and Constrained Neural Acceleration**

2.1 Main Research Question

How can one generate fast, stable, and financially consistent American option price and Greek surfaces while preserving the obstacle constraint and accurately recovering the early-exercise boundary?

2.2 Concrete Subquestions

1. How do volatility, interest rate, dividend yield, and maturity affect the early-exercise boundary?
2. Where do Delta and Gamma become unstable near the exercise boundary?
3. Can a constrained neural surrogate reproduce price, Greek, and boundary behaviour without violating payoff, monotonicity, or convexity constraints?
4. In which regimes is the neural surrogate useful as an accelerator, and in which regimes should one rely directly on the classical solver?

2.3 Minimum Publishable Contribution

The minimum successful contribution is a validated computational framework that produces:

- American put and dividend-paying American call prices;
- early-exercise boundary curves;
- Delta and Gamma surfaces;
- stress-regime risk maps over (r, q, σ, T) ;
- diagnostic checks for obstacle violation, monotonicity, convexity, and numerical stability.

The neural-network component is valuable only if it improves fast evaluation without destroying financial consistency. It is an accelerator and diagnostic tool, not the backbone of the project.

3 Literature Review: What Students Need to Know

Students should read the literature in four layers. The aim is not to collect many papers, but to understand where the project fits.

3.1 Layer 1: Classical American Option Pricing

American options can be formulated as optimal stopping problems, free-boundary PDE problems, variational inequalities, or linear complementarity problems (LCPs). Classical methods remain essential because they provide stable low-dimensional benchmarks.

- Brennan and Schwartz developed finite-difference methods for American option valuation and helped establish the PDE/free-boundary numerical tradition [1].
- Longstaff and Schwartz introduced least-squares Monte Carlo (LSM), which made American-style option valuation practical in path-dependent and higher-dimensional settings [2].
- Haugh and Kogan developed duality methods that can produce upper bounds for American option prices [3].

For this project, the classical method is not merely a baseline. It is the reference standard used to generate trusted labels, boundaries, and Greeks.

3.2 Layer 2: Scientific Machine Learning for PDEs

Physics-informed and PDE-based neural methods attempt to solve differential equations by embedding the governing equation into the training loss.

- Physics-informed neural networks (PINNs) were popularized by Raissi, Perdikaris, and Karniadakis [4].
- The Deep Galerkin Method (DGM) proposed a mesh-free deep learning approach for high-dimensional PDEs [5].

These methods are attractive for financial PDEs, but students must be cautious: training can be unstable, constraint enforcement can be weak, and impressive plots may hide poor behaviour near exercise boundaries.

3.3 Layer 3: Deep Learning for American Options

There are two main neural-network traditions for American options.

Optimal stopping methods. These methods learn an exercise policy or continuation value. Becker, Cheridito, and Jentzen proposed deep optimal stopping methods [6], and later work applied related ideas to pricing and hedging American-style options [7].

Free-boundary and PINN methods. Recent papers use neural networks to represent the price, the free boundary, or both. Meshless PINN methods have been applied to American options [8]. Nwankwo, Umeorah, Ware, and Dai proposed a deep-learning free-boundary framework using a Landau transformation and boundary-aware neural construction [9]. Other recent work has explored fractional Black–Scholes free-boundary problems [10] and jump-diffusion process-informed neural networks for American options [11].

This literature implies an important strategic conclusion: a plain PINN for an American put is no longer a strong contribution. The project must instead focus on risk surfaces, boundary recovery, Greek stability, and financial consistency.

3.4 Layer 4: Application Direction

The application angle is computational risk analytics. In practice, a desk or risk system does not need one isolated price. It needs reliable surfaces:

- price as a function of spot, time, volatility, rate, and dividend yield;
- Delta and Gamma for hedging;
- early-exercise boundary for exercise policy interpretation;
- stress maps showing how the boundary changes under market regimes.

This is where the project can be distinctive. The paper should show that the free-boundary structure is not a mathematical detail, but a central object in American option risk management.

4 Mathematical Formulation

4.1 Risk-Neutral Model

Let S_t be the underlying asset price. Under the risk-neutral measure, assume the dividend-adjusted Black–Scholes dynamics

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t, \quad (1)$$

where r is the risk-free rate, q is the continuous dividend yield, and σ is volatility.

The value of an American option with maturity T and payoff $\Phi(S)$ is

$$V(S, t) = \sup_{\tau \in [t, T]} \mathbb{E} \left[e^{-r(\tau-t)} \Phi(S_\tau) \mid S_t = S \right]. \quad (2)$$

For a put and a call,

$$\Phi_{\text{put}}(S) = (K - S)^+, \quad \Phi_{\text{call}}(S) = (S - K)^+. \quad (3)$$

4.2 Variational Inequality

Define the Black–Scholes spatial operator

$$\mathcal{L}V = \frac{1}{2}\sigma^2 S^2 V_{SS} + (r - q)SV_S - rV. \quad (4)$$

In original time t , the American option satisfies the complementarity conditions

$$V(S, t) \geq \Phi(S), \quad (5)$$

$$-(V_t + \mathcal{L}V) \geq 0, \quad (6)$$

$$(V(S, t) - \Phi(S))[-(V_t + \mathcal{L}V)] = 0. \quad (7)$$

In the continuation region, $V > \Phi$ and the Black–Scholes PDE holds. In the exercise region, $V = \Phi$.

4.3 Forward Time Variable

For numerical work, use time-to-maturity

$$\tau = T - t. \quad (8)$$

Then $U(S, \tau) = V(S, T - \tau)$ satisfies, in the continuation region,

$$U_\tau = \mathcal{L}U. \quad (9)$$

The initial condition is the payoff:

$$U(S, 0) = \Phi(S). \quad (10)$$

The LCP form in forward time is

$$U(S, \tau) \geq \Phi(S), \quad (11)$$

$$U_\tau - \mathcal{L}U \geq 0, \quad (12)$$

$$(U(S, \tau) - \Phi(S))(U_\tau - \mathcal{L}U) = 0. \quad (13)$$

5 Numerical Benchmark: CN/PSOR

5.1 Why This Solver Comes First

Before building neural networks, students must produce a reliable benchmark. The CN/PSOR solver is used for three reasons:

1. Crank–Nicolson gives a second-order time-centred discretisation in the continuation region.
2. PSOR directly enforces the obstacle constraint.
3. The method produces interpretable boundary and Greek information.

5.2 Grid

Choose a truncated domain $S \in [0, S_{\max}]$ and time grid

$$S_i = i\Delta S, \quad i = 0, \dots, M, \quad \tau_n = n\Delta\tau, \quad n = 0, \dots, N. \quad (14)$$

Use $S_{\max}$ large enough that the boundary condition does not affect the region of interest. A starting choice is

$$S_{\max} = 4K \quad (15)$$

for the Black–Scholes experiments, followed by sensitivity checks.

5.3 Discrete Operator

At an interior node S_i , approximate

$$U_S(S_i, \tau_n) \approx \frac{U_{i+1}^n - U_{i-1}^n}{2\Delta S}, \quad (16)$$

$$U_{SS}(S_i, \tau_n) \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{(\Delta S)^2}. \quad (17)$$

Let L_h denote the resulting finite-difference matrix for $\mathcal{L}$ on the interior nodes.

5.4 Crank–Nicolson LCP

From τ_n to τ_{n+1} , the continuation equation is approximated by

$$\left(I - \frac{\Delta\tau}{2}L_h\right)U^{n+1} = \left(I + \frac{\Delta\tau}{2}L_h\right)U^n. \quad (18)$$

For the American option, solve at each time level the LCP

$$AU^{n+1} \geq b^n, \quad (19)$$

$$U^{n+1} \geq \Phi, \quad (20)$$

$$(U^{n+1} - \Phi)^\top (AU^{n+1} - b^n) = 0, \quad (21)$$

where

$$A = I - \frac{\Delta\tau}{2}L_h, \quad b^n = \left(I + \frac{\Delta\tau}{2}L_h\right)U^n, \quad (22)$$

with boundary adjustments included in b^n .

5.5 Boundary Conditions

For an American put, use

$$U(0, \tau) = K, \quad U(S_{\max}, \tau) \approx 0. \quad (23)$$

For a dividend-paying American call, use

$$U(0, \tau) = 0, \quad U(S_{\max}, \tau) \approx \max\{S_{\max} - K, S_{\max}e^{-q\tau} - Ke^{-r\tau}\}. \quad (24)$$

This upper condition blends the immediate-exercise asymptotic value with the large-spot European asymptotic value. It is especially useful for checking the case $q = 0$, where an American call should match its European counterpart. The upper boundary should still be checked by increasing $S_{\max}$. If the numerical results in the region $S/K \in [0.4, 1.8]$ change materially when $S_{\max}$ changes, the domain is too small.

5.6 Validation Ladder

Students should validate the solver in the following order:

1. **European closed-form test.** Disable the projection step and compare the finite-difference price with the Black–Scholes closed form.
2. **No-dividend American call test.** With $q = 0$, the American call should agree with the European call, because early exercise is not optimal.
3. **American put reference test.** Compare selected put values against a high-resolution grid or a trusted reference table.
4. **Grid refinement test.** Refine both ΔS and $\Delta\tau$ and check price, boundary, and Greek stability.
5. **Domain sensitivity test.** Increase $S_{\max}$ and verify that the region of interest is unchanged.

This ladder should be completed before any large parameter sweep.

5.7 Rannacher Smoothing

The terminal payoff has a kink at $S = K$, and Crank–Nicolson can produce oscillatory Greeks near non-smooth data. For production experiments, use Rannacher smoothing: replace the first one or two Crank–Nicolson time steps by several backward Euler half-steps. Then continue with Crank–Nicolson. The paper should report whether smoothing was used, because it affects Delta and Gamma near maturity.

5.8 PSOR Iteration

At each time step, solve the LCP using projected successive over-relaxation. Let x denote the vector of unknown option values at the new time level. For row i of $Ax = b$, define the Gauss–Seidel candidate

$$y_i = \frac{1}{A_{ii}} \left(b_i - \sum_{j < i} A_{ij} x_j^{\text{new}} - \sum_{j > i} A_{ij} x_j^{\text{old}} \right). \quad (25)$$

The projected SOR update is

$$x_i^{\text{new}} = \max \left\{ \Phi_i, x_i^{\text{old}} + \omega(y_i - x_i^{\text{old}}) \right\}. \quad (26)$$

Use $\omega \in (1, 2)$ and tune it empirically. Stop when

$$\|x^{k+1} - x^k\|_{\infty} < \varepsilon. \quad (27)$$

5.9 Boundary Extraction

After solving for $U(S, \tau)$, define the continuation premium

$$P(S, \tau) = U(S, \tau) - \Phi(S). \quad (28)$$

The exercise boundary is the transition between $P = 0$ and $P > 0$. If S_i is the last exercise node and S_{i+1} is the first continuation node, estimate

$$S_f(\tau) \approx S_i - \frac{P_i}{P_{i+1} - P_i} (S_{i+1} - S_i). \quad (29)$$

For puts, the exercise region is typically low S . For dividend-paying calls, the exercise region is typically high S when dividends are large enough.

5.10 Greeks

Compute Delta and Gamma from the numerical surface:

$$\Delta_i^n \approx \frac{U_{i+1}^n - U_{i-1}^n}{2\Delta S}, \quad (30)$$

$$\Gamma_i^n \approx \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{(\Delta S)^2}. \quad (31)$$

Near the payoff kink and near the free boundary, Gamma can be unstable. This is not a bug by itself; it is part of the mathematical difficulty. The question is whether the numerical method produces controlled and interpretable Greek surfaces.

6 Application Experiments

6.1 Base Parameter Grid

Use the following grid for the first complete study:

Parameter	Values
S/K range	0.4 to 1.8
T	0.25, 0.5, 1.0, 2.0
σ	0.2, 0.4, 0.6
r	0.01, 0.05, 0.10
q	0.00, 0.03, 0.06, 0.10

The strike should be normalised to $K = 1$ for experiments unless market data are introduced later.

6.2 Required Figures

Students should produce the following figures before starting the neural surrogate:

1. American put price surface over (S, τ) .
2. Dividend-paying American call price surface over (S, τ) .
3. Put early-exercise boundary for different σ .
4. Call early-exercise boundary for different q .
5. Delta surface and Gamma surface for one put and one dividend-paying call.
6. Heatmap of early-exercise boundary location as a function of (q, σ) .
7. Grid-convergence plot for price and boundary.
8. Runtime versus grid size.

6.3 Required Diagnostics

For every experiment, report:

$$\text{ObstacleViolation} = \max_{i,n} (\Phi_i - U_i^n)^+, \quad (32)$$

$$\text{ComplementarityError} = \max_{i,n \geq 1} |(U_i^n - \Phi_i)(AU^n - b^{n-1})_i|. \quad (33)$$

The complementarity residual is evaluated only on interior nodes and only after the first time step, because b^{n-1} is the right-hand side used to compute U^n .

Also check shape constraints:

- put price should be decreasing in S ;
- call price should be increasing in S ;
- option prices should be convex in S ;
- the American value should dominate the European value.

6.4 Expected Qualitative Checks

The following checks are not formal proofs, but they are useful for catching implementation errors:

- for American puts, higher volatility usually lowers the exercise boundary because continuation value increases;
- for dividend-paying American calls, higher dividend yield usually expands the early-exercise region;
- as maturity approaches, the exercise boundary should approach the payoff kink;
- the no-dividend American call should show no genuine early-exercise boundary.

7 Neural Surrogate Stage

7.1 When to Start

Students should begin this stage only after the CN/PSOR benchmark passes the diagnostics above. A neural model trained on an unreliable solver simply learns unreliable labels.

7.2 Training Data

Generate labels from the benchmark solver. Use normalised inputs:

$$x = (\log(S/K), \tau/T, r, q, \sigma, T, \text{type}). \quad (34)$$

Suggested outputs are:

$$y = (U/K, \Delta, K\Gamma, P/K), \quad (35)$$

where $P = U - \Phi$ is the continuation premium.

Do not train heavily on Gamma labels exactly at the payoff kink or exactly on the extracted boundary. Those locations are mathematically delicate and can dominate the loss in an unhelpful way. Use a mask that downweights or excludes Gamma labels in a narrow band around $S = K$ and around $S_f(\tau)$, while still reporting error there as a stress test.

7.3 Model Variants

Train three models:

1. **Plain surrogate:** price-only supervised neural network.
2. **Greek-aware surrogate:** supervised network trained on price, Delta, and Gamma labels.
3. **Constrained surrogate:** Greek-aware surrogate plus financial penalties.

7.4 Constraint Penalties

Let $\widehat{U}_\theta$ be the neural price. The constrained loss should have the form

$$\begin{aligned} \mathcal{J}(\theta) = & \lambda_p \|\widehat{U}_\theta - U\|^2 + \lambda_\Delta \|\widehat{\Delta}_\theta - \Delta\|^2 + \lambda_\Gamma \|\widehat{\Gamma}_\theta - \Gamma\|^2 \\ & + \lambda_{\text{obs}} \left\| (\Phi - \widehat{U}_\theta)^+ \right\|^2 + \lambda_{\text{shape}} \mathcal{J}_{\text{shape}} + \lambda_{\text{pde}} \mathcal{J}_{\text{pde}}. \end{aligned} \quad (36)$$

The shape penalty should enforce:

$$\hat{U}_S \leq 0 \quad \text{for puts,} \quad (37)$$

$$\hat{U}_S \geq 0 \quad \text{for calls,} \quad (38)$$

$$\hat{U}_{SS} \geq 0 \quad \text{for both puts and calls.} \quad (39)$$

The PDE penalty should be applied primarily in the continuation region, where the benchmark premium P is safely positive. Do not force the continuation PDE inside the exercise region.

7.5 Train-Test Split

Do not use only a random split over grid points. That can make the model appear better than it is. Use both:

- random point split for interpolation diagnostics;
- regime split, where entire parameter combinations are held out.

The regime split is the more important test.

7.6 Success Criteria

The neural surrogate is useful only if it satisfies all of the following:

1. small price error on held-out regimes;
2. lower obstacle and shape violations than the plain surrogate;
3. acceptable Delta and Gamma error near the continuation region;
4. much faster evaluation than recomputing a full CN/PSOR grid;
5. no misleading boundary behaviour.

If these criteria are not met, keep the neural model as an exploratory appendix or remove it from the paper.

7.7 Ablation Table

The neural section should include an ablation table with at least the following rows:

Model	Price error	Delta error	Constraint violation	Query time
Plain surrogate				
Greek-aware surrogate				
Constrained surrogate				

If the constrained model does not improve constraint violation or regime generalisation, do not claim that SciML improves the application.

8 Step-by-Step Implementation Plan

8.1 Suggested Repository Structure

```
american-risk-surfaces/  
  src/  
    solvers/  
      cn_psor.py  
      boundaries.py  
      greeks.py  
    experiments/  
      run_grid.py  
      convergence.py  
      diagnostics.py  
    models/  
      surrogate.py  
      losses.py  
    plotting/  
      figures.py  
  configs/  
    base_put.yaml  
    base_call_dividend.yaml  
  notebooks/  
    01_solver_validation.ipynb  
    02_boundary_and_greeks.ipynb  
    03_neural_surrogate.ipynb  
  results/  
  paper/
```

8.2 Week-by-Week Plan

Week	Tasks and deliverables
1	Implement clean finite-difference solver and validate the European closed-form case. Then add projection for American puts and obstacle/complementarity diagnostics.
2	Add dividend-paying American calls. Verify that the no-dividend American call matches the European call. Check boundary conditions and domain sensitivity.
3	Implement Delta, Gamma, Rannacher smoothing, and boundary extraction. Add grid-convergence tests and runtime tables.
4	Run application stress experiments over (r, q, σ, T) . Produce risk maps and decide which figures are paper-worthy.
5	Generate supervised training data from the validated solver. Build the plain neural surrogate.
6	Add Greek-aware labels and shape/obstacle penalties. Compare plain and constrained surrogates.
7	Run held-out-regime tests. Stress-test boundary and Greek behaviour. Decide whether neural results belong in the main paper.
8	Assemble paper draft: methodology, experiments, limitations, and reproducible code notes.

9 Decision Gates

9.1 Gate 1: Solver Reliability

Proceed only if:

- European closed-form tests pass;
- the no-dividend American call agrees with the European call;
- obstacle violation is near machine/tolerance level;
- price convergence improves under grid refinement;
- boundary curves are stable when the grid is refined;
- boundary curves are stable when $S_{\max}$ is increased.

9.2 Gate 2: Application Strength

Proceed to neural modelling only if the application study already has clear messages. For example:

- higher dividends push dividend-paying calls toward earlier exercise;
- higher volatility changes the exercise boundary and increases Greek sensitivity near the boundary;
- low-rate regimes produce qualitatively different exercise behaviour;
- Gamma instability is localised near the exercise frontier and payoff kink.

9.3 Gate 3: Neural Surrogate Inclusion

Include the neural surrogate in the main paper only if it adds a genuine contribution. It must be faster and financially more consistent than a plain black-box network. Otherwise, the paper should remain a strong computational finance application paper based on the classical solver.

10 Expected Paper Structure

1. Introduction: American option risk surfaces and why the free boundary matters.
2. Literature review: classical free-boundary methods, optimal stopping, and SciML for option pricing.
3. Mathematical formulation: VI/LCP form of American options.
4. Benchmark method: CN/PSOR, boundary extraction, Greeks, diagnostics.
5. Application experiments: stress-regime risk maps.
6. Neural surrogate: model, constraints, training, held-out tests.
7. Results: price, boundary, Greek, violation, and runtime comparisons.
8. Limitations and future work: Heston, jump diffusion, rough volatility, and market calibration.

11 Target Journal Strategy

The most realistic first target is *Applied Mathematics and Computation*, provided the paper emphasises a computational method with application-driven numerical results. A simulation-heavy version could target *Mathematics and Computers in Simulation*. A more numerical-analysis-heavy version could target *Applied Numerical Mathematics*, but that version would require stronger analysis and convergence discussion.

The paper should not be submitted to a finance-facing journal unless real market data or a sharper economic insight is added.

12 Common Mistakes to Avoid

1. Do not write a broad paper claiming that neural networks replace finite differences.
2. Do not compare methods without a financial question.
3. Do not train a neural network before the solver is validated.
4. Do not report only price error; Greeks and boundary behaviour are central to the project.
5. Do not force the PDE residual inside the exercise region.
6. Do not use a random train-test split as the only generalisation test.
7. Do not hide failed regimes. They are often the most informative part of the paper.

13 Student Checklist

Before reporting results to the supervisor, students should be able to answer:

1. What is the exact PDE or LCP being solved?
2. What are the boundary conditions, and have they been tested?
3. What is the PSOR tolerance and relaxation parameter?
4. How does the solution change under grid refinement?
5. Where is the early-exercise boundary, and how is it extracted?
6. Are Delta and Gamma computed in a stable way?
7. Are obstacle, monotonicity, and convexity constraints satisfied?
8. Which parameter regimes are financially interesting?
9. What does the neural surrogate add beyond speed?
10. Which result would remain if the neural network section were removed?

14 Supervisor’s Bottom Line

The project is feasible if it stays focused. The strongest route is:

Build a rigorous free-boundary benchmark, use it to produce American option risk surfaces under stress regimes, and add constrained neural acceleration only after the numerical foundation is trustworthy.

If students follow this order, the project can become a credible Q1-oriented computational finance paper. If they start from an unfocused PINN comparison, the project will likely become too broad and too fragile.

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